Mouse DEA April 2026

Ishaan Banwait

Sample Preparation (TransBIOTech)

The ARDS model was induced by intranasal administration of LPS and mice were treated with the different test products 1h before LPS exposure.

Intranasal RoA is through nose to target lungs directly. These groups (1-5) were discontinued due to mice dying, with remaining samples spread across the **intraperitoneal** RoA: in abdominal cavity (groups 6-10).

Vehicle = kolliphor HS 15

CFA and CFB diluted in DMSO. DEX diluted in water.

The mice were treated with the different test products 1h before LPS administration.

Groups	Model	Treatment	Dose	n
6	Saline	Vehicle	0	5
7	LPS	Vehicle	0	13
8	LPS	CFA	2.5 mg/kg	13
9	LPS	CFB	2.5 mg/kg	12
10	LPS	Dexamethasone	2.5 mg/kg	13

Phenotypic Observations

Mice respiratory functions was measured before and 24h after LPS exposure using whole-body plethysmography.

RNA Extraction

24h after LPS administration, mice were euthanized by cardiac puncture and tissue was extracted from the right lung and preserved in RNAlater.

Vehicle = kolliphor HS 15.

All LPS models: mice were treated with the different test products 1h before LPS administration. RNA was collected 24h post LPS.

RNA Sequencing (GeneMarkers)

- Lipopolysaccharide (LPS)-induced lung injury mouse model tissues received on 23 August, 2023
- RNA was extracted from mouse lung tissue using a Maxwell RSC SimplyRNA kit
- Libraries were prepared using the Illumina Stranded RNA kit which uses PolyA enrichment to remove rRNA

- Sequencing was performed on an Illumina NextSeq 550Dx platform
- Total of 12 biological samples (4 replicates x CFA, CFB, control)
- Single-ended sequencing was performed, producing an R1 file for each sample

TransBIOTech Mouse ID	Treatment	GeneMarkers Sample IDs (H23-)
1897	Vehicle	142
1899	Vehicle	143
1901	Vehicle	144
1972	Vehicle	145
1902	Cannflavin A	146
1903	Cannflavin A	147
1904	Cannflavin A	148
1974	Cannflavin A	149
1909	Cannflavin B	150
1910	Cannflavin B	151
1913	Cannflavin B	152
1914	Cannflavin B	153

Count Generation (Gurkamal)

These files were received from gurkamal@canurta.com on March 22nd in a private Teams chat.

Step	Tool	Input	Output	Notes
QC	FastQC + MultiQC	Raw reads (12 R1 files)	Assessments of read quality, adapter content, duplication, and GC distributions multiqc_report.html	
Trimming	fastp	Raw reads (12 FASTQ files)	Trimmed reads (12 FASTQ files)	
Alignment	STAR	Raw reads (12 FASTQ files) Reference genome	12 SAM/BAM files 12 Log.final.out files	Downloaded the mm9 reference genome and the complementary ensemble GTF (annotation). STAR aligns each read.
Quantification	featureCounts	12 BAM files	featureCounts_raw.txt featureCounts_raw.txt.summary	

Parsing	Pandas	featureCounts_raw.txt	gene_counts_matrix.tsv (genes x samples)	Removed the extra columns from the featureCounts table and renamed the sample columns from BAM file paths to sample IDs so the count matrix could be used in DESeq2 (genes x samples). This needs to be transposed (samples x genes) for pydeseq2.

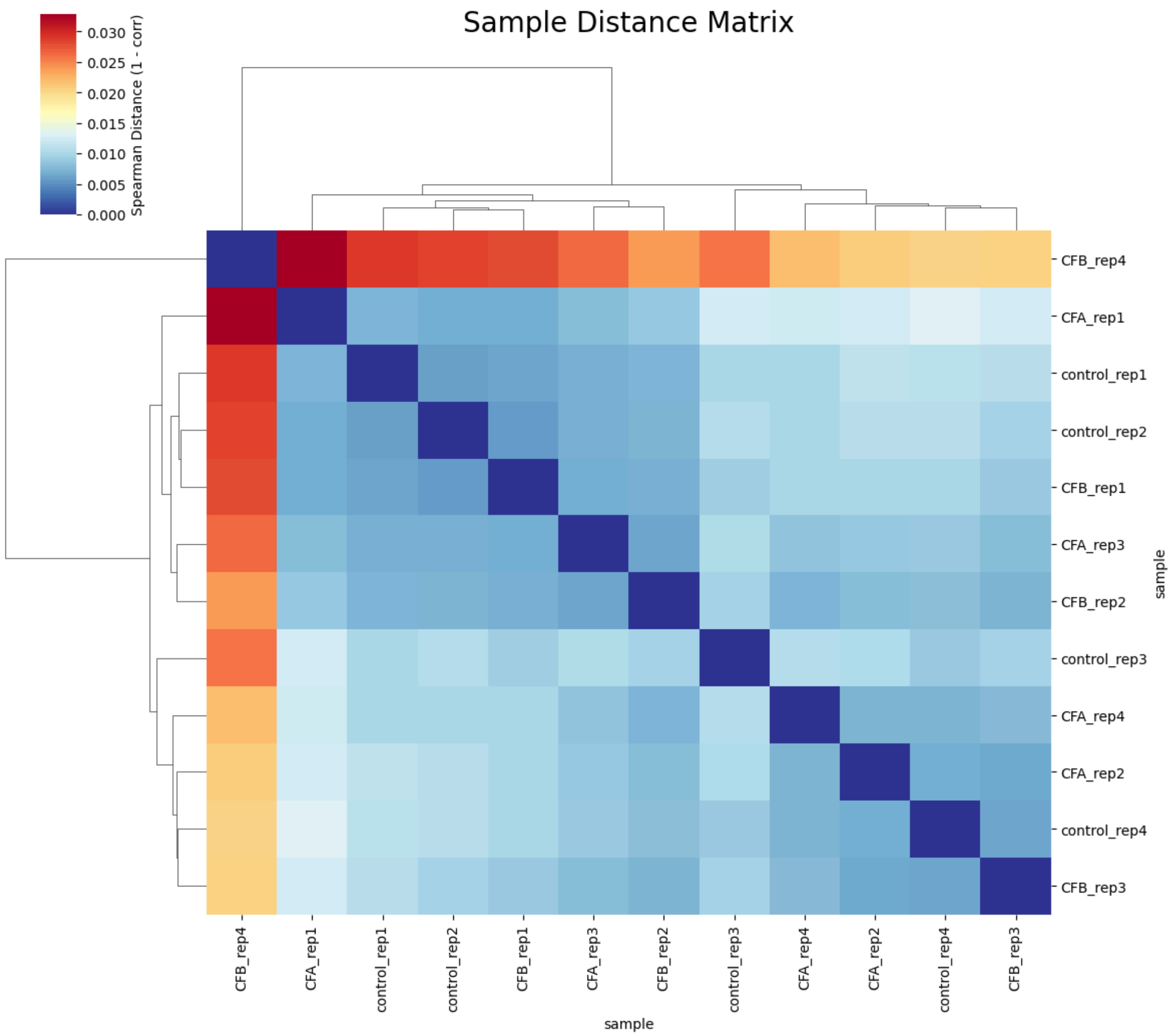
Sample-Level Quality Control

Sample Screening

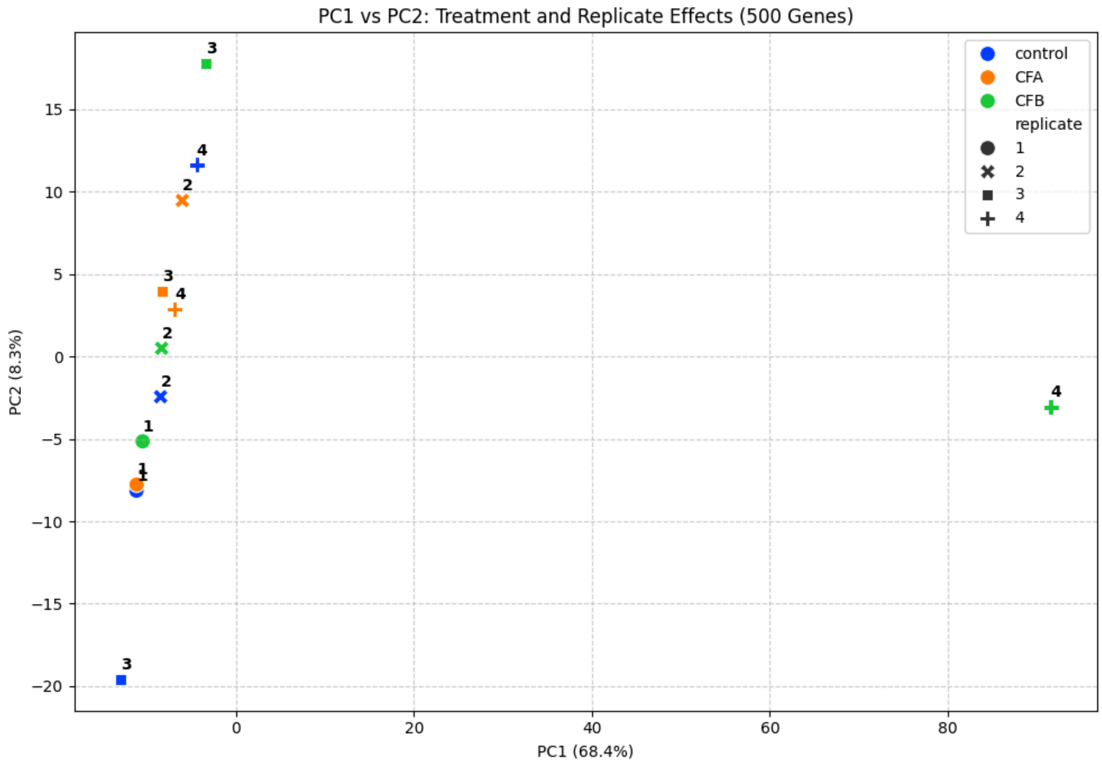
Every sample passed absolute quality control thresholds. Signals were consistent, all within 2 MADs from the sample mean.

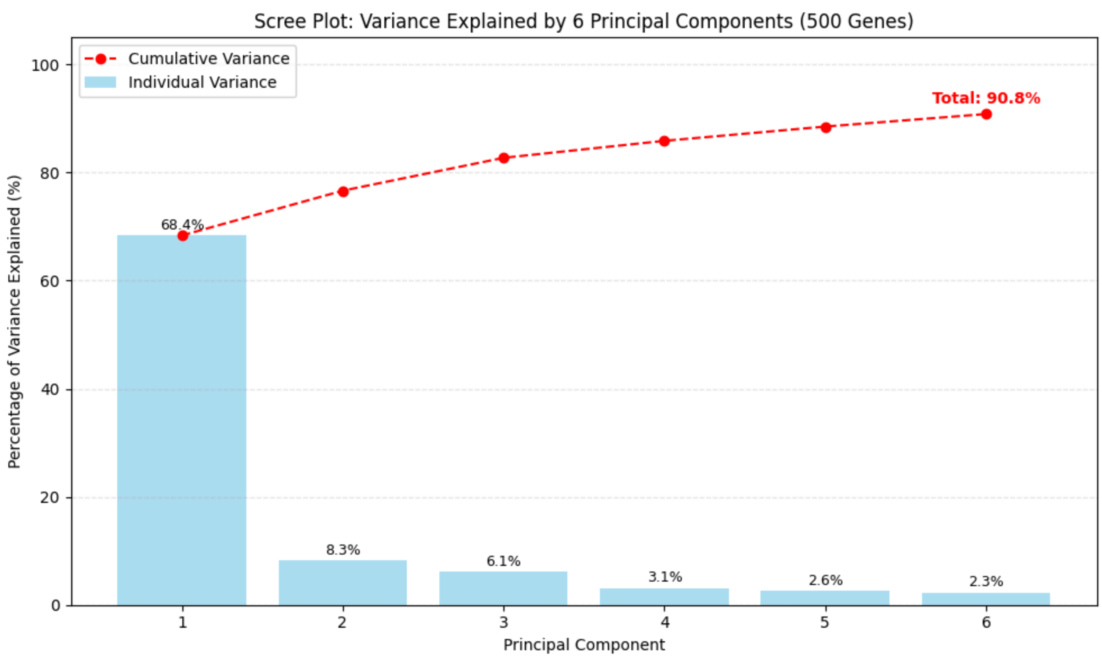
Sample Outlier Detection

A sample correlation heatmap of all samples shows chaotic clustering, with CFB_rep4 standing out as a clear outlier.

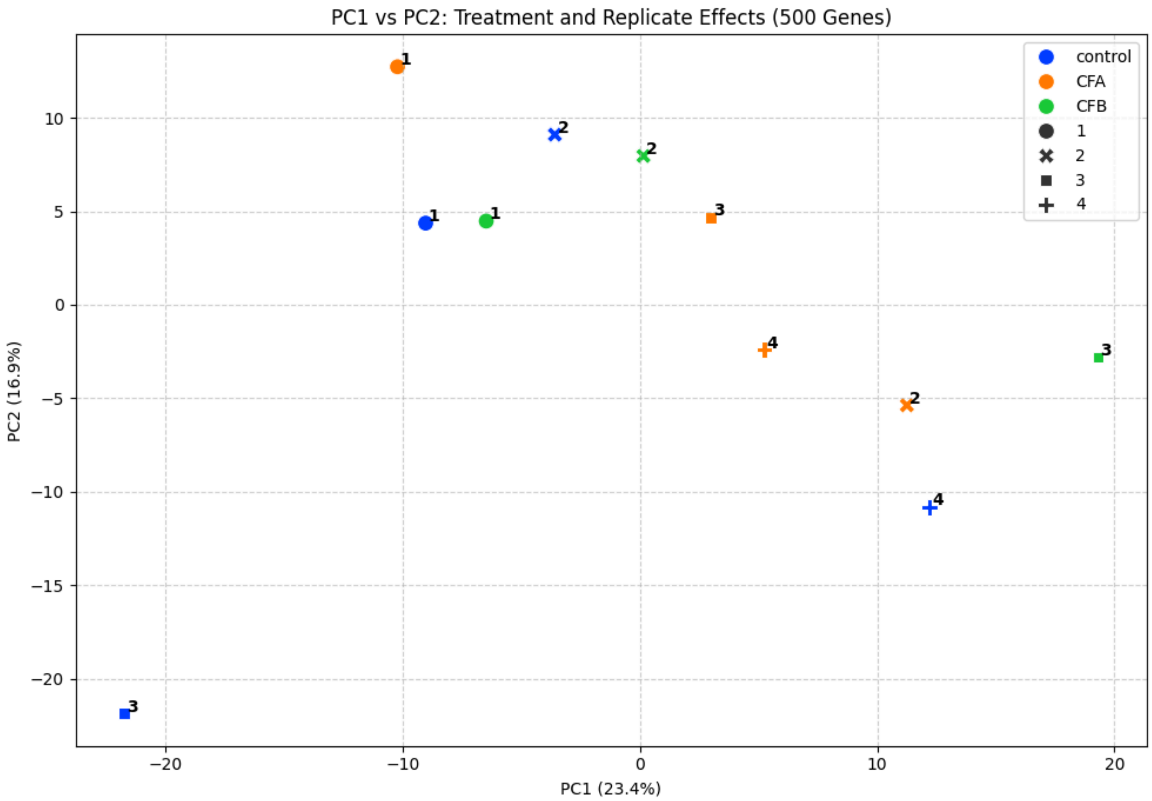


CFB_rep4 is dominating the PCA, standing out as PC1 representing 68.4% of the variance. It must be removed. It also appears that control_rep 3 may be an outlier on PC2 representing 8.3% of the variance.



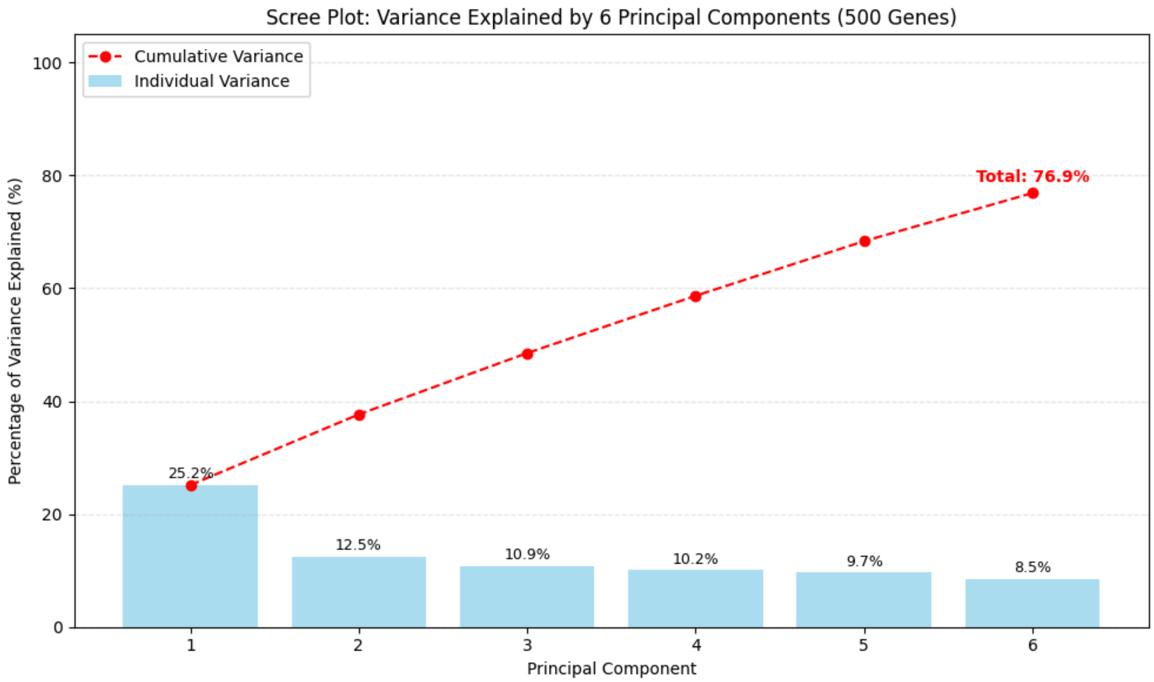


After removing CFB_rep4, PC1 (23.4%) and PC2 (16.9%) represent another outlier: control_rep3.

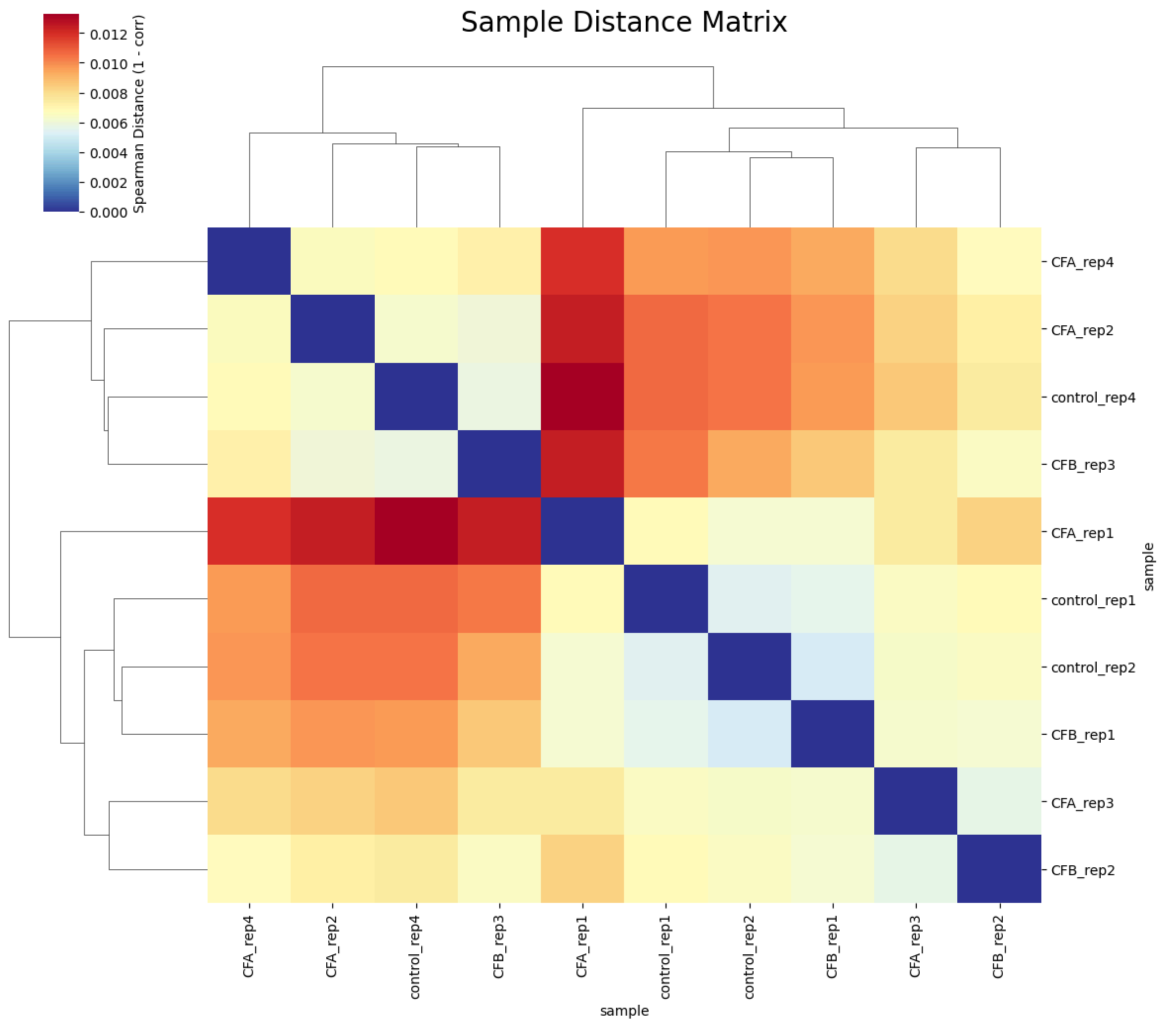


Post-Filtering

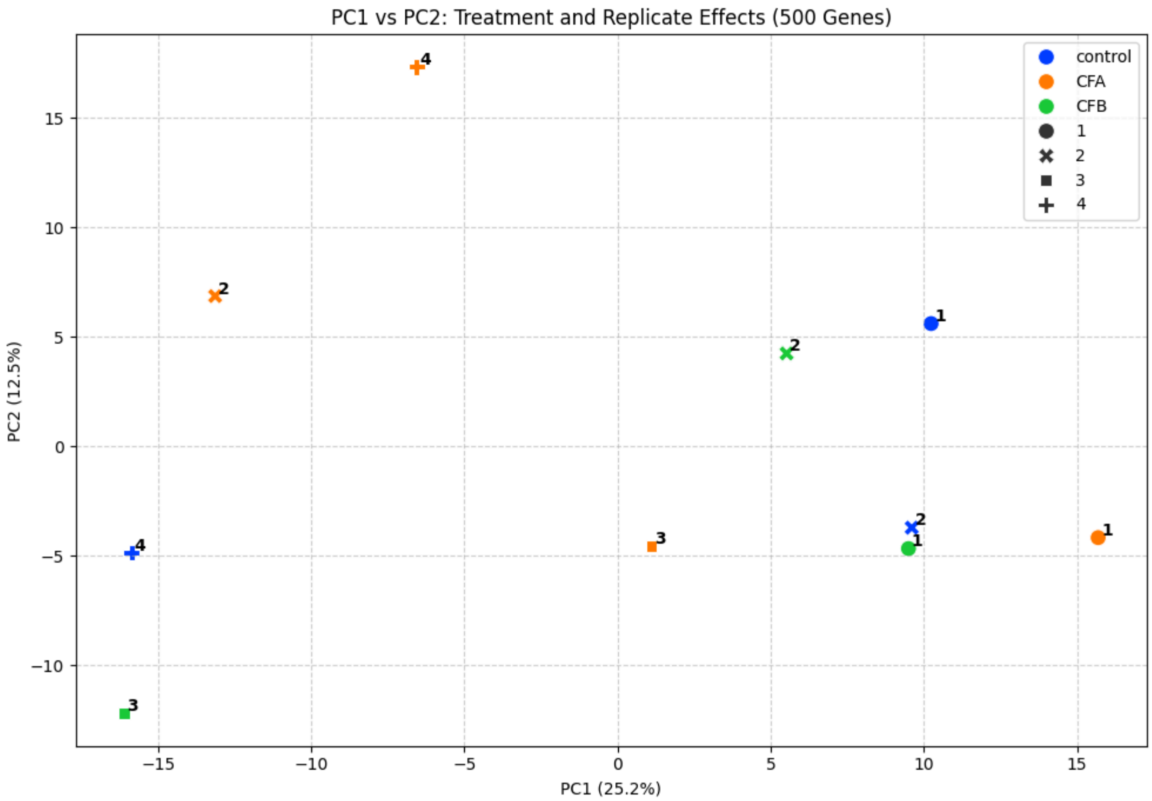
After removing both CFB_rep4 and control_rep3, clustering still look messy. Pattern is the same with the top 500 most-variable, top 100 most-variable, and all 37992 genes.



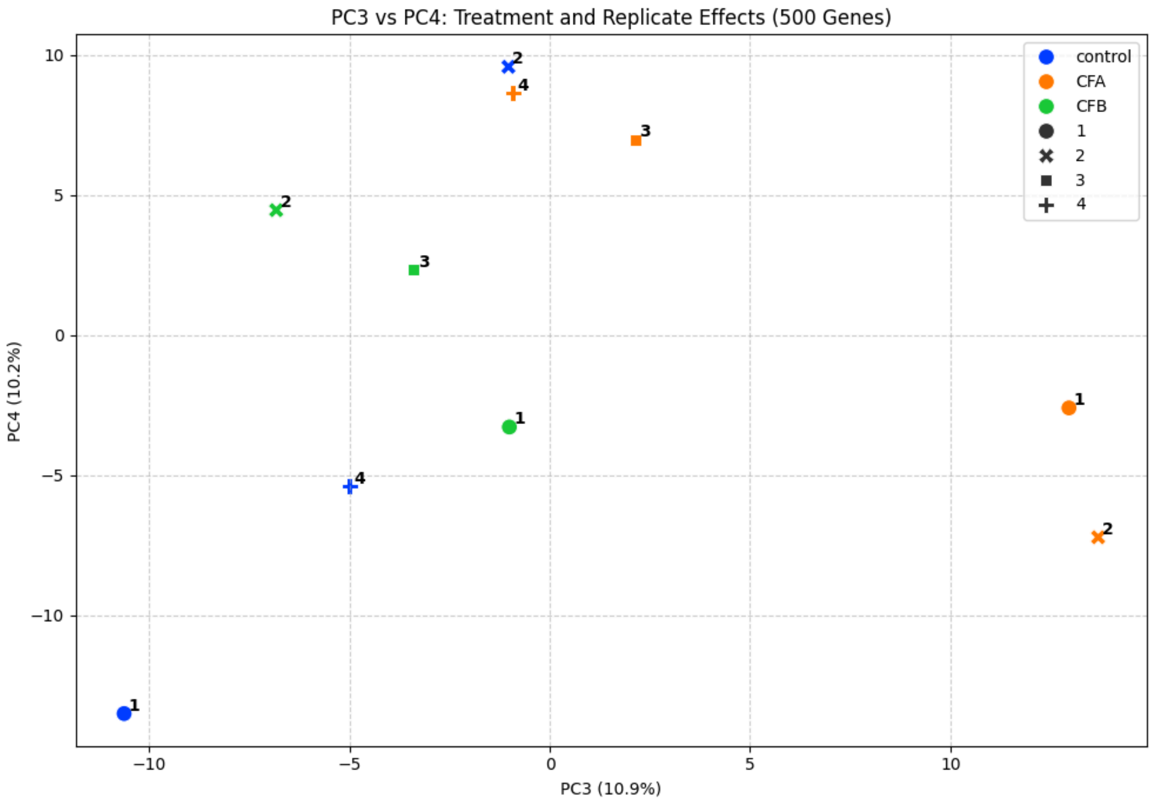
Sample clustering heatmap pattern is very similar with Pearson, Spearman, and Kendall.

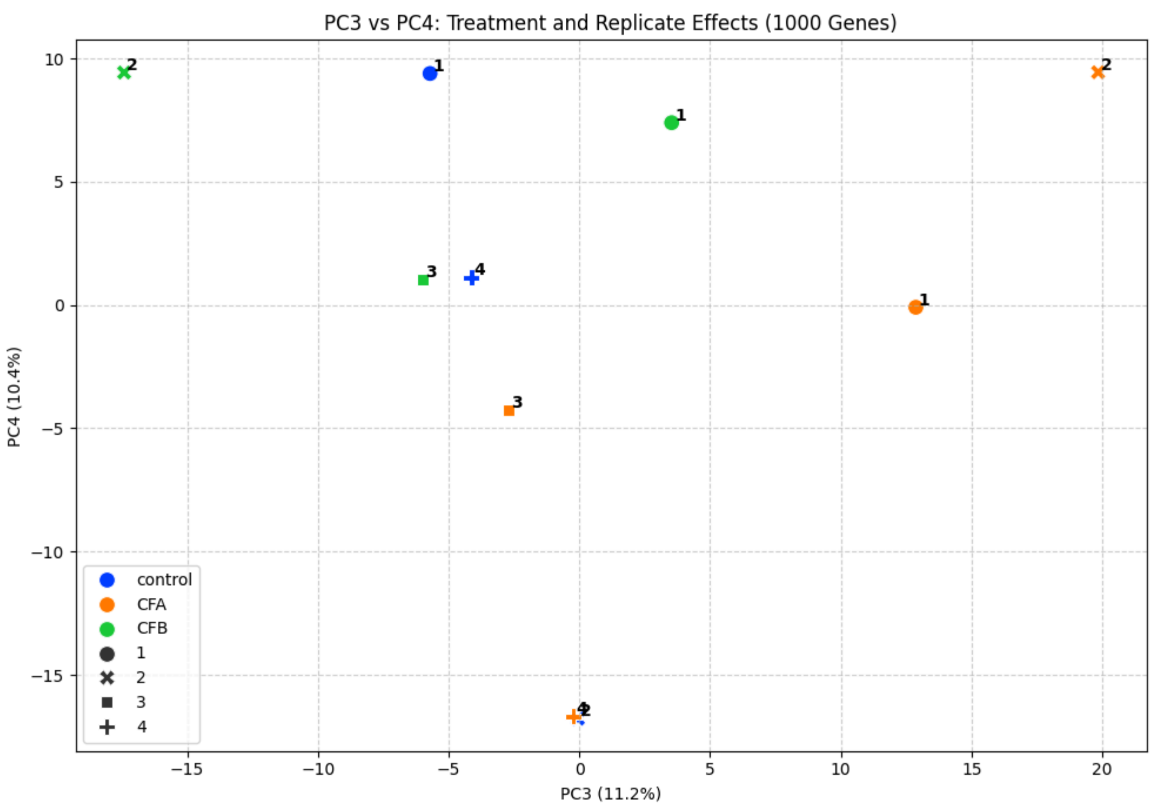


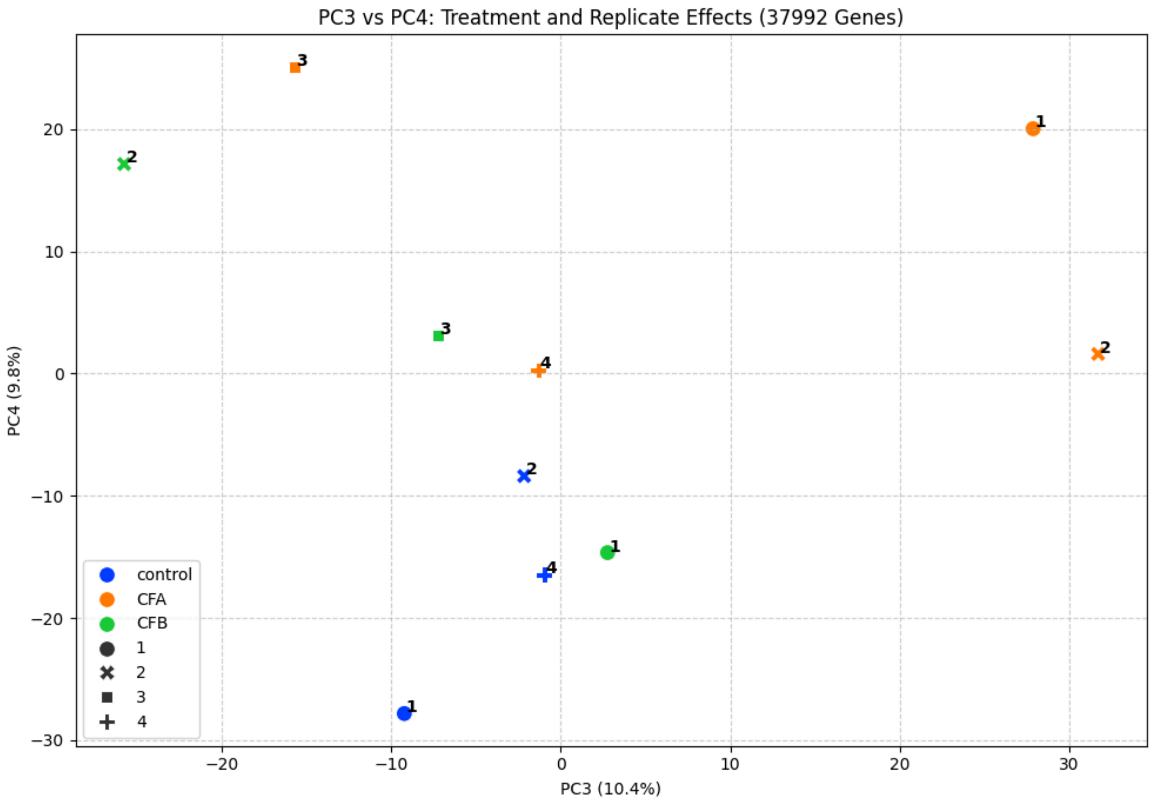
PC1 and PC2 could be explanatory variables; PC1 most likely being sex.



Treatment effects could potentially be on PC3, where CFA is showing separation from negative control. This is not bad if PC1 and PC2 can be controlled.


PC3 vs PC4: Treatment and Replicate Effects (500 Genes)
control
CFA
CFB
1
2
3
4
PC3 (10.9%)
PC4 (10.2%)


PC3 vs PC4: Treatment and Replicate Effects (1000 Genes)
control
CFA
CFB
1
2
3
4
PC3 (11.2%)
PC4 (10.4%)



Differential Expression Analysis

Gene and Sample Filtering

10 samples entered the statistical analysis after filtering out the two outliers above. After filtering out genes with less than 10 counts, 18,902 of a total 37,992 were retained.

Metadata dataframe that entered the `DeseqDataSet`.

sample	sample_ID	treatment	replicate
control_rep1	H23-142	control	1
control_rep2	H23-143	control	2
control_rep4	H23-145	control	4
CFA_rep1	H23-146	CFA	1
CFA_rep2	H23-147	CFA	2
CFA_rep3	H23-148	CFA	3
CFA_rep4	H23-149	CFA	4
CFB_rep1	H23-150	CFB	1
CFB_rep2	H23-151	CFB	2
CFB_rep3	H23-152	CFB	3

Statistical Design

- A single `DeseqDataSet` was fit, so that there was one global GLM per gene for all 10 samples. This allows for more statistical power when estimating gene-wide dispersions since more replicates are available, helping with controlling the noise in this messy dataset.
- A single-factor (treatment) reference-means model was used with control as the reference level (intercept; β_0). The other two treatments were each given a coefficient referenced to this intercept (β_2, β_3). This setup aligns with the Drosophila analysis where results were in the form of treatment tested against a pathological control as reference.

The design matrix used in this analysis.

sample	**Intercept (β_0)**	**treatment[T.CFA] (β_1)**	**treatment[T.CFB] (β_2)**
control_rep1	1.0	0	0
control_rep2	1.0	0	0
control_rep4	1.0	0	0
CFA_rep1	1.0	1	0
CFA_rep2	1.0	1	0
CFA_rep3	1.0	1	0
CFA_rep4	1.0	1	0
CFB_rep1	1.0	0	1
CFB_rep2	1.0	0	1
CFB_rep3	1.0	0	1

The contrasts used in this analysis, of format ['variable_of_interest', 'tested_level', 'ref_level'].

`["treatment", "CFA", "control"]`

`["treatment", "CFB", "control"]`